

DHL Group

Dangerous Goods Handling Manual

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1. Scope & Applicability

This manual covers the handling, storage, and transport of dangerous goods across all DHL business units. It is mandatory for all employees who accept, handle, store, or transport dangerous goods in any DHL facility or vehicle.

DHL classifies dangerous goods according to the UN Recommendations on the Transport of Dangerous Goods. All 9 hazard classes are covered in this manual, with specific procedures for the most commonly handled classes in DHL operations.

2. Classification & Identification

2.1 UN Number Verification: Every dangerous goods shipment must be identified by its 4-digit UN number. The shipper is responsible for correct classification, but DHL staff must verify the UN number against the declared description before acceptance.

2.2 Hazard Classes Commonly Handled by DHL:

Class 1: Explosives (restricted - DHL Express does not accept; DHL Freight by special arrangement only)

Class 2: Gases (compressed, liquefied, dissolved) - e.g., aerosol cans, fire extinguishers, CO2 cartridges

Class 3: Flammable liquids - e.g., paints, adhesives, perfumes, fuel samples. Flash point must be declared.

Class 4: Flammable solids - e.g., matches, metal powders, self-heating substances

Class 5: Oxidizers & organic peroxides - e.g., hydrogen peroxide, bleach

Class 6: Toxic & infectious substances - e.g., pesticides, clinical waste, diagnostic specimens

Class 7: Radioactive materials (restricted - special license required per country)

Class 8: Corrosives - e.g., batteries (sulfuric acid), cleaning chemicals, mercury

Class 9: Miscellaneous - e.g., lithium batteries, dry ice, environmentally hazardous substances

3. Acceptance Procedures

3.1 Shipper Declaration: A completed Shipper Declaration for Dangerous Goods (SDDG) is mandatory for all DG shipments by air. For road/sea, the transport document per ADR/IMDG must accompany the shipment.

3.2 Acceptance Checklist: Before accepting any DG shipment, the trained DG acceptance officer must verify: (a) correct and complete documentation, (b) proper packaging per packing instruction, (c) correct marks and labels, (d) no damage or leakage, (e) quantity limits per package and per vehicle/aircraft not exceeded.

3.3 Rejection Criteria: Reject and document any shipment that: shows signs of leakage or damage, has incomplete/incorrect documentation, exceeds quantity limits, uses non-UN-certified packaging for goods requiring it, or is offered by an untrained shipper.

4. Storage Requirements

4.1 Segregation Rules: Dangerous goods must be segregated according to the compatibility matrix in Appendix A. Key rules: (a) oxidizers must be separated from flammable materials by minimum 3 meters or a fire-resistant barrier, (b) acids and alkalis must not be stored together, (c) lithium batteries must be stored in a dedicated area with sand extinguishers.

4.2 Lithium Battery Storage: Given the increasing volume of lithium battery shipments, DHL has established enhanced storage requirements: maximum 500 kg net weight per storage unit, temperature monitoring (alarm at >35C), fire

detection with thermal cameras, dedicated fire suppression (water mist + inert gas), monthly inspection of stored items for swelling or damage.

4.3 Temperature-Controlled DG: Certain dangerous goods (e.g., organic peroxides Class 5.2) require temperature-controlled storage. The required control temperature and emergency temperature must be monitored continuously. Alert thresholds are set 5C below the emergency temperature.

5. Emergency Response

5.1 Spill Response: All DG storage areas must have spill kits appropriate to the hazard classes stored. Absorbent materials for liquid spills, neutralizing agents for acids/alkalis, and sealed disposal containers must be available within 10 meters of any DG storage location.

5.2 Lithium Battery Fire Protocol: In case of a lithium battery thermal event: (a) evacuate the immediate area (50m radius), (b) do NOT use CO2 or dry powder extinguishers, (c) use water mist or sand, (d) call the fire brigade immediately and inform them of lithium battery involvement, (e) do not attempt to move burning batteries.

5.3 Emergency Contacts: DHL Global Emergency Hotline: +49-228-XXX-XXXX (24/7). Local emergency services: posted at every DG storage area entrance.

6. Training Requirements

6.1 Mandatory Training: All employees handling dangerous goods must complete DG awareness training before first assignment. Role-specific training levels: (a) Awareness - all warehouse staff (4 hours, online), (b) Function-specific - acceptance officers, drivers, loaders (16 hours, classroom + practical), (c) Specialist - DG Safety Advisors (40 hours, certified by authority).

6.2 Recurrent Training: All DG training must be refreshed every 24 months. For air transport, recurrent training is required every 24 months per IATA DGR. Records must be kept for minimum 3 years after last training date.

6.3 Competency Verification: After training completion, all staff must pass a written exam (minimum 80% score) and a practical assessment for function-specific and specialist levels.